

Quantifying Uncertainty

QBA 5330 – Module 2, Part 1 (October 16)

QBA 5330

Where We're Headed Today

- **Part 1:** Quantifying uncertainty, from probability to sampling distributions
- **Part 2:** Confidence intervals for proportions and means
- **Part 3:** Introduction to hypothesis testing, and launching the Module 2 deliverable

Today's Business Decision

A retailer has piloted a new online storefront in a sample of markets. Should it launch nationally?

We'll carry this same decision through all three parts.

A Quick Example

Leadership estimates:

- National launch cost: **\$2,000,000**
- Expected incremental profit: **\$2,600,000**
- Standard deviation of that profit estimate: **\$1,800,000**
- **Is this a “safe bet”?**

Recap: Expected Value and Risk

- Expected value is a **long-run average**, not a guarantee for any one outcome
- A positive expected value can still carry real risk
- We need both the **center** (expected value) and the **spread** (standard deviation) to reason about risk

Recap: Sampling Distributions

- A sample mean or sample proportion is a **statistic**
- Every statistic has a **sampling distribution**: how much it would vary across repeated samples
- The **standard error** is the standard deviation of that sampling distribution

The Two Standard Error Formulas

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}} \qquad SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

A bigger sample doesn't eliminate uncertainty. It **shrinks the standard error**.

The Pilot Data

- **Conversion:** $n = 200$ visitors, 30 converted, $\hat{p} = 0.15$
- **Order value:** $n = 25$ orders, $\bar{x} = \$71.50$, $s = \$42$

(You've seen these numbers before, in your Week 4–6 reading.)

Your Turn (20 minutes)

In your group, compute both standard errors:

```
1 p_hat <- 0.15; n_conv <- 200
```

```
1 s <- 42; n_order <- 25
```

Then discuss: what does each number mean for the launch decision?

Before the Break

We now have:

- A point estimate ($\hat{p} = 0.15, \bar{x} = \71.50)
- A sense of how much each could bounce around (the standard errors)
- **What we don't have yet:** a defensible *range* to bring to leadership.
- That's next.